LAURA NGO

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Summary

Data Analyst with 3 years of experience using Python, R, and SQL to transform complex business data into clear insights. Builds dashboards and recurring reports for business stakeholders, investigates data discrepancies, and improves data reliability through validation and reconciliation. Experienced in translating analytical deliverables and automating manual, spreadsheet-based workflows.

Experience

- **Compass Education** — Los Angeles, CA
  Administrative Coordinator (Data & Mailing Operations) — *August 2024 – Present*
  - Analyze **12,000+** biweekly and monthly mail-volume records across 10+ client groups to identify recurring and seasonal demand patterns, supporting forecasting and resource planning for leadership.
  - Develop and maintain recurring dashboards and operational reports for leadership, delivered on biweekly and monthly cycles, giving stakeholders visibility into volume trends, workload changes, and resource needs.
  - Investigate and flag data discrepancies, work with cross-functional teams to identify root causes, and explain changes in results to stakeholders; implemented process improvements that strengthened data reliability and workflow efficiency.
- **UCLA Education & Information Studies** — Los Angeles, CA
  Data Analysis Research Assistant — *July 2024 – August 2024*
  - Digitized **10,000+** paper records into structured datasets for a decade-long K–12 study, and maintained a Google Sheets reporting infrastructure supporting analysis of **500+** datasets, accelerating delivery of research insights by **40%**.
  - Performed data validation, exploratory data analysis, statistical modeling, and result interpretation to identify patterns, assess reliability, and communicate findings.

Skills

- **Analytics & Querying:** SQL (PostgreSQL), Python, Pandas, NumPy, R, Exploratory Data Analysis, Statistical Modeling, Regression, Variance Analysis, Forecasting
- **Dashboards & Reporting:** Tableau, Excel, Google Sheets, HTML/CSS/JavaScript, Recurring Executive Reporting, Self-Serve Reporting
- **Data Quality & Engineering:** Data Cleaning, Data Validation, Data Reconciliation, ETL Pipelines, PySpark, Parquet, DVC
- **Tools:** Jupyter, Google Colab, GitHub, VS Code, RStudio, DBeaver, UCSD Expanse
- **Professional:** Cross-functional Collaboration, Stakeholder Communication, Data-Driven Decision-Making, Adaptability

Projects

- **Steam Player Retention & Pipeline Analysis** *(PySpark, Python, DVC, Parquet, Expanse)*
  - Built an end-to-end analytics and machine learning pipeline processing 1.2M+ player profiles and 43GB+ of behavioral data, including ingestion, quality auditing, transformation, and feature engineering to analyze player retention.
  - Implemented data validation checks throughout the pipeline using Parquet, Spark, DVC, and UCSD Expanse for reproducible workflows; published an interactive website presenting findings to technical and non-technical audiences.
- **Amazon Reviews Rating Prediction & Dashboard** *(PySpark, Python, Machine Learning)*
  - Processed **102.9M** Amazon reviews (117GB) in PySpark: cleaned and filtered data, engineered features, and trained and tuned Random Forest models to predict star ratings.
  - Diagnosed severe class imbalance (60M 5-star vs. 5.2M 1-star reviews) through exploratory analysis and confusion-matrix review.
- **Earthquake Activity & Seismic Monitoring Analysis** *(JavaScript, D3.js, HTML/CSS, Geospatial Visualization)*
  - Developed interactive global earthquake visualizations using earthquake and seismic station data to explore geographic patterns in seismic activity and monitoring coverage.
  - Created an animated California earthquake heatmap to examine spatial concentrations and temporal variation in earthquake activity, presenting findings through an interactive website.
- **COVID-19 Deaths & Poverty Regression Analysis** *(R, R Markdown)*
  - Examined associations between county-level poverty rates, socioeconomic conditions, and COVID-19 mortality to investigate geographic disparities in pandemic outcomes.
  - Developed reproducible R Markdown analyses of 2,634 county-level observations across 53 variables, applying descriptive statistics, correlation analysis, and regression modeling to evaluate relationships among variables.

Education

- **Master of Data Science**, *University of California, San Diego* — La Jolla, CA
- **Bachelor of Statistics & Data Science**, *University of California, Los Angeles* — Los Angeles, CA